

Duke Faculty Survey: Written Report

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Introduction

This project uses a probability sample to study faculty represented in 17 departments in Duke University's Trinity College of Arts & Sciences. The first three questions concern the proportion who are women, the proportion with at least one grant ongoing in 2026, and the average number of years since receiving a Ph.D. among those who hold one. The fourth asks what proportion of Ph.D. holders received their degree from Duke University.

Project code is available at <https://github.com/yliclaire/duke-faculty-survey>.

Sampling design

The sampling frame came from department-filtered Faculty results on Duke Scholars and contained 1,152 unique faculty across 17 departments. Faculty with multiple affiliations appeared once and were assigned to the alphabetically first included department before outcomes were collected. The target population is the faculty represented in this frame. Because Scholars only approximates departmental rosters, incomplete or incorrect affiliations could cause coverage error.

We used proportionate stratified simple random sampling without replacement, with department as the stratum. The total sample was fixed at 230, about 20% of the frame. We allocated selections by department size using the largest-remainder method, then sampled independently within each department with a fixed random seed.

Stratification ensured representation from all 17 departments and could improve precision because the proportion of women may vary by department. We did not use cluster sampling because reviewing individual profiles was the main collection cost, and we lacked an appropriate auxiliary size variable for probability-proportional-to-size sampling.

In department h , a faculty member's inclusion probability was $\pi_{hi} = n_h/N_h$, where N_h is the department frame size and n_h is its sample size. The survey weight was $w_{hi} = 1/\pi_{hi} = N_h/n_h$. Because the allocation was approximately proportional, weights were close to 5.

Data collection

For each sampled faculty member, we collected information from Duke Scholars about whether the person was a woman, grants, and Ph.D. education. A faculty member was classified as having an ongoing grant when at least one listed grant, in any role, had dates that included 2026. No listed grant was coded as no ongoing grant.

A person counted as a Ph.D. holder only when the education section explicitly listed a Ph.D. or D.Phil. When available, we recorded the degree year and institution and calculated years since the degree as 2026 – Ph.D. year. The fourth outcome indicated whether a Ph.D. holder received that degree from Duke.

We first reviewed gendered pronouns on each profile to determine whether a faculty member was a woman; this classified 95 faculty. When a profile offered no usable pronoun evidence, we used the first name as an initial aid if U.S. Social Security records for 1930–2005 associated it with one sex at least 95% of the time. We manually reviewed the remaining 28 cases, including uncertain or conflicting evidence and one pronoun match that referred to someone else. Each sampled faculty member received a final classification from the profile information, although names, pronouns, and profile text can all lead to error.

Analysis

We used the `survey` package in R, specifying department as the stratum, inverse inclusion probabilities as weights, and department frame sizes as finite population corrections. Questions 1 and 2 used survey-weighted means of binary indicators. For Questions 3 and 4, we treated Ph.D. holders as a domain by subsetting the survey design. Question 4 used a logit confidence interval to keep its bounds between zero and one. All intervals have 95% confidence. Code is also available at github.com/yliclaire/duke-faculty-survey.

Results

1. **Faculty who are women.** The estimated proportion is 41.5% (95% CI: 36.3% to 46.6%).
2. **Faculty with an ongoing grant in 2026.** An estimated 31.9% have at least one ongoing grant in 2026 (95% CI: 27.2% to 36.7%).
3. **Years since receiving a Ph.D.** Among the 186 faculty identified as Ph.D. holders, the estimated mean is 27.8 years (95% CI: 25.8 to 29.8 years).
4. **Ph.D. holders who received the degree from Duke.** Among the 186 identified Ph.D. holders, the estimated proportion is 10.7% (95% logit CI: 7.4% to 15.2%).

Limitations

The estimates describe the 17-department frame and may not generalize to excluded or unlisted Trinity faculty. Scholars profiles may be incomplete or outdated, particularly for grants and education. Coding no listed grant as no ongoing grant may miss unlisted awards. Determining whether a faculty member was a woman required judgment, and missing Ph.D. years or institutions could affect Questions 3 and 4. The confidence intervals do not account for these sources of measurement and coverage error.